⇒ Gini - Impurity Index :

$$Gini(Situation) = 1 - \sum_{i=1}^{c} (p_i)^2$$

Outlook	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes
Sunny	Mild	Normal	True	Yes
Overcast	Mild	High	True	Yes
Overcast	Hot	Normal	False	Yes
Rainy	Mild	High	True	No

$$Gini(9,5) = 1 - \left\{ \left(\frac{9}{14}\right)^2 + \left(\frac{5}{14}\right)^2 \right\}$$

$$= 0.46$$

$$Gini(PlayTennis, Outlook)$$

$$= P(Sunny) \times G(3,2) + P(Overcast) \times G(4,0) + P(Rainy) \times G(2,3)$$

$$= \left\{ \left(\frac{5}{14}\right) \times 0.48 + \left(\frac{4}{14}\right) \times 0 + \left(\frac{5}{14}\right) \times 0.48 \right\}$$

$$= 0.343$$

Gini (PT, Temp) = 0.444

Gini (PT, Humidity) = 0.3675

Gini (PT, Windy) = 0.428

Decision Tree Problems:

-> Overfitting & Underfitting

-> Depth of Tree.

Ensemble Models: Combination of multiple non-parametric models.

i) Boosting: Amplification of parameters.

ii) Bagging: Aggregation of parameters.

(Bootstrap Aggregation)

(Bootstrap Aggregation)

Bagging: Random Forest

